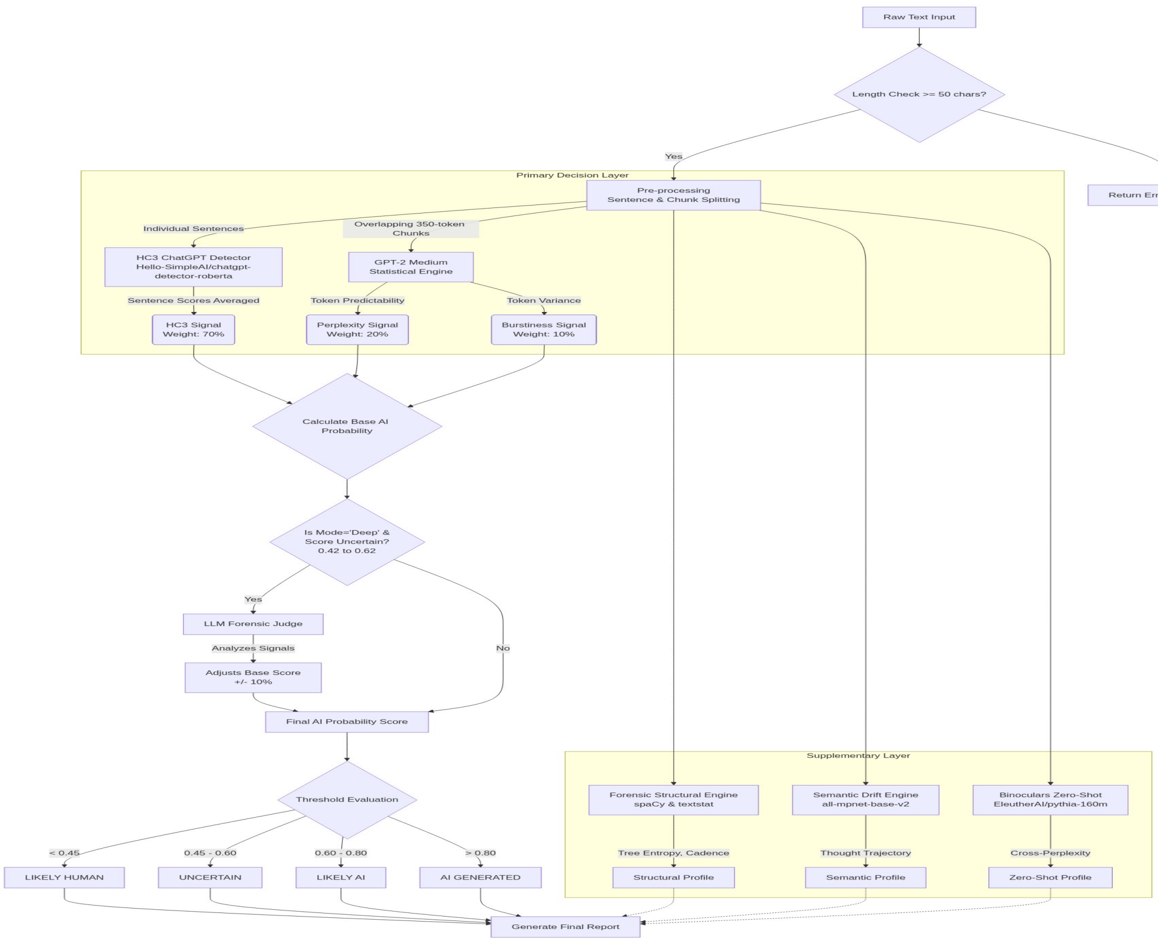
FakeShield AI Text Lab:
Architecture & Processing Pipeline

The AI Text Lab is a high-performance forensic engine (currently version 14.0) designed to distinguish between human-written and AI-generated text. It employs a multi-layered approach that combines state-of-the-art transformer models for direct classification with statistical analysis and linguistic theory to form a robust, tamper-resistant verdict.

1. Text Processing Flowchart

The following flowchart illustrates the lifecycle of a text query as it passes through the various forensic engines.



2. Detailed Pipeline Breakdown

Phase 1: Pre-processing

When text is submitted, the engine first ensures it meets a minimum length requirement (50 characters). To avoid the standard 512-token limit of BERT-based models, which causes truncation and inaccurate scoring on long documents, the text is split in two ways:

1. **Sentence Array:** Split into individual sentences for the HC3 classifier.
2. **Overlapping Chunks:** Split into 350-token chunks with a 200-token stride for statistical analysis.

Phase 2: The Core Ensemble (Probability Scoring)

The final AI probability score is calculated exclusively using a weighted ensemble of the following two models:

A. HC3 ChatGPT Detector (70% Weight)

Model: Hello-SimpleAI/chatgpt-detector-roberta
Mechanism: This is a RoBERTa-based model trained on the Human-ChatGPT Comparison Corpus (HC3).
Approach: It runs inference on every single sentence individually. The engine then calculates a weighted average of these sentence scores, giving higher importance to sentences the model is highly confident about. This sentence-average architecture prevents long documents from "hiding" AI-generated paragraphs at the end of the text.

B. GPT-2 Statistical Engine (30% Weight)

Model: gpt2-medium
Mechanism: AI models generate text by selecting the most statistically probable next word. Human writing is much less predictable. This engine measures two metrics across the overlapping chunks:

> **Perplexity (20% Weight):** Measures token predictability. Low perplexity (<32) is a strong indicator of AI generation. High perplexity (>55) indicates human writing.
>
> **Burstiness (10% Weight):** Measures the variance in token probability. AI text tends to have a uniform, flat probability distribution (low burstiness). Human writing features irregular sentence structures and varying vocabulary, leading to 'bursts' of complexity (high burstiness).

Phase 3: Supplementary Forensic Profiling

Simultaneously, the text is processed by supplementary engines. These do not alter the main AI probability score but provide crucial explainability and linguistic metrics for the user interface.

C. Forensic Structural Engine

Models: spaCy (en_core_web_sm) and textstat
Mechanism: Analyzes the dependency tree of the sentences.
Philosophy: AI models are trained to produce perfectly balanced, grammatically "clean" syntax trees (high uniformity). Authentic human writing is often structurally "lopsided" and irregular. It calculates depth variance, punctuation randomness, and sentence length cadence.

D. Semantic Drift Engine

Model: SentenceTransformer (all-mpnet-base-v2)
Mechanism: Analyzes the "Thought Flow Trajectory".
Philosophy: AI writing follows a highly logical, smooth, "geodesic" path from one concept to the next. Human reasoning is naturally associative, frequently featuring "jumps" and topic deviations. This engine measures semantic entropy and drift variance.

E. Binoculars Zero-Shot Engine

Models: EleutherAI/pythia-160m (Observer) and pythia-160m-deduped (Performer)
Mechanism: Computes the log-probability ratio between the two models. If both models agree on the text, it is likely machine-generated. If the observer is surprised but the performer isn't, it indicates human authorship.

Phase 4: Tie-Breaker (Deep Mode)

If the analysis is running in "deep" mode and the ensemble score falls into the "Uncertain" threshold (42% to 62% AI probability), the **Forensic Judge** is invoked.

Mechanism: An LLM prompt call acts as an expert judge. It reviews the raw text alongside the layer scores (HC3, Perplexity, Burstiness) to provide a final ruling.
Impact: The judge can bump the final score up or down by a maximum of 10% and extracts specific "suspicious indicators" for the final report.

3. Verdict Thresholds

The final probability score determines the threat level and verdict:

Probability Score	Verdict	Threat Level
0.00 to 0.44	LIKELY HUMAN	LOW
0.45 to 0.59	UNCERTAIN	MEDIUM
0.60 to 0.79	LIKELY AI	HIGH
0.80 to 1.00	AI GENERATED	CRITICAL